Matrices & Vectors

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and B be the matrix with rows $\langle 2, -1\rangle$ and $\langle 0, 5\rangle$. Find $A + B$ and $2A$.

$A + B$ has rows $\langle 3, 2\rangle$ and $\langle -2, 9\rangle$. $2A$ has rows $\langle 2, 6\rangle$ and $\langle -4, 8\rangle$.

2 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 1, 3\rangle$, $\langle -2, 4\rangle$) and B (rows $\langle 2, -1\rangle$, $\langle 0, 5\rangle$) from above, find AB.

AB has rows $\langle 2, 14\rangle$ and $\langle -4, 22\rangle$: e.g. row 1, column 1 is $(1)(2) + (3)(0) = 2$.

3 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and let $\mathbf{v} = \langle 3, -1\rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (1)(3) + (3)(-1),\ (-2)(3) + (4)(-1)\rangle = \langle 0, -10\rangle$.

4 **[Foundational]** • **Target: Determinant and inverse**

Find $\det A$ for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$, and state whether A is invertible.

$\det A = (1)(4) - (3)(-2) = 4 + 6 = 10 \neq 0$, so A is invertible.

5 **[Foundational]** • **Target: Determinant and inverse**

Find A^{-1} for the matrix A with rows $\langle 1, 3\rangle$ and $\langle -2, 4\rangle$.

$A^{-1} = \frac{1}{10} \times$ the matrix with rows $\langle 4, -3\rangle$ and $\langle 2, 1\rangle$, i.e., A^{-1} has rows $\langle 0.4, -0.3\rangle$ and $\langle 0.2, 0.1\rangle$.

6 **[Foundational]** • **Target: Solving systems with the inverse matrix**

Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).

A^{-1} has rows $\langle 0.4, -0.3\rangle$ and $\langle 0.2, 0.1\rangle$. Then $x = 0.4(13) - 0.3(4) = 5.2 - 1.2 = 4$ and $y = 0.2(13) + 0.1(4) = 2.6 + 0.4 = 3$. Check: $4 + 3(3) = 13$ and $-2(4) + 4(3) = 4$.

7 **[Foundational]** • **Target: Vector magnitude and dot product**

For $\mathbf{u} = \langle 4, 3\rangle$ and $\mathbf{w} = \langle -3, 4\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$\mathbf{u} \cdot \mathbf{w} = (4)(-3) + (3)(4) = 0$, and $|\mathbf{u}| = \sqrt{4^2 + 3^2} = 5$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Angle between vectors**

Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.

$\mathbf{p} \cdot \mathbf{q} = 6$, $|\mathbf{p}| = 2\sqrt{2}$, $|\mathbf{q}| = 3$. $\cos\theta = \frac{6}{6\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ$.

9 **[Intermediate]** • **Target: Angle between vectors**

Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.

Dot product $= 14 - 3 = 11$, magnitudes $\sqrt{50}$ and $\sqrt{13}$. $\cos\theta = \frac{11}{\sqrt{650}} \approx 0.4314 \Rightarrow \theta \approx 64.4^\circ$.

10 **[Intermediate]** • **Target: 3D Vectors & Cross Products**

Find the angle θ between $\mathbf{u} = \langle -6, -7 \rangle$ and $\mathbf{v} = \langle 2, 6 \rangle$, to the nearest 0.1°.

$\mathbf{u} \cdot \mathbf{v} = -54$, $|\mathbf{u}| = \sqrt{85}$, $|\mathbf{v}| = \sqrt{40}$. $\cos\theta = \frac{-54}{\sqrt{85}\sqrt{40}} \approx -0.9261$, so $\theta \approx 157.8^\circ$ (Theorem 2.2).

11 **[Intermediate]** • **Target: Addition and scalar multiplication**

Find the angle θ between $\mathbf{u} = \langle 6, 8 \rangle$ and $\mathbf{v} = \langle 7, -4 \rangle$, to the nearest 0.1°.

$\mathbf{u} \cdot \mathbf{v} = 10$, $|\mathbf{u}| = \sqrt{100}$, $|\mathbf{v}| = \sqrt{65}$. $\cos\theta = \frac{10}{\sqrt{100}\sqrt{65}} \approx 0.1240$, so $\theta \approx 82.9^\circ$ (Theorem 2.2).

12 **[Intermediate]** • **Target: Matrix multiplication**

Find the angle θ between $\mathbf{u} = \langle 6, -7 \rangle$ and $\mathbf{v} = \langle -3, 0 \rangle$, to the nearest 0.1°.

$\mathbf{u} \cdot \mathbf{v} = -18$, $|\mathbf{u}| = \sqrt{85}$, $|\mathbf{v}| = \sqrt{9}$. $\cos\theta = \frac{-18}{\sqrt{85}\sqrt{9}} \approx -0.6508$, so $\theta \approx 130.6^\circ$ (Theorem 2.2).

13 **[Intermediate]** • **Target: Determinant and inverse**

Find the angle θ between $\mathbf{u} = \langle -5, 5 \rangle$ and $\mathbf{v} = \langle -3, 8 \rangle$, to the nearest 0.1°.

$\mathbf{u} \cdot \mathbf{v} = 55$, $|\mathbf{u}| = \sqrt{50}$, $|\mathbf{v}| = \sqrt{73}$. $\cos\theta = \frac{55}{\sqrt{50}\sqrt{73}} \approx 0.9104$, so $\theta \approx 24.4^\circ$ (Theorem 2.2).

14 **[Intermediate]** • **Target: Solving systems with the inverse matrix**

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Vector magnitude and dot product • Cross-Lesson: 3.01 Functions & Graphs

Find the product of the roots of $x^2 + 14x + 113 = 0$.

By Vieta's formulas, the product of the roots of $x^2 + Bx + C = 0$ is C; here that's 113 (the imaginary parts of the conjugate pair cancel in the sum, and multiply to a real number in the product).

16 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Introductory Calculus

Find the product of the roots of $x^2 - 18x + 106 = 0$.

By Vieta's formulas, the product of the roots of $x^2 + Bx + C = 0$ is C; here that's 106 (the imaginary parts of the conjugate pair cancel in the sum, and multiply to a real number in the product).

17 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 6.01 Probability

Let $z = 3 - 4i$. What is the real part of z?

A complex number is generally expressed in the form $a + bi$, where a is the real part and b is the imaginary part. In the given complex number $z = 3 - 4i$, the real part is the term that does not have i multiplied by it. Therefore, the real part of z is 3.

18 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

Find A^{-1} for $A = [-3, -6; -5, -6]$.

$\det A = -12$. Swap the diagonal entries, negate the off-diagonal entries, divide by $\det A$: $A^{-1} = \left[\frac{1}{2}, -\frac{1}{2}; -\frac{5}{12}, \frac{1}{4}\right]$. Final Answer: $\left[\frac{1}{2}, -\frac{1}{2}; -\frac{5}{12}, \frac{1}{4}\right]$

19 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

Find A^{-1} for $A = [2, -5; -1, -2]$.

$\det A = -9$. Swap the diagonal entries, negate the off-diagonal entries, divide by $\det A$: $A^{-1} = \left[\frac{2}{9}, -\frac{5}{9}; -\frac{1}{9}, -\frac{2}{9}\right]$. Final Answer: $\left[\frac{2}{9}, -\frac{5}{9}; -\frac{1}{9}, -\frac{2}{9}\right]$

20 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Does $y = 6x^2 + 9$ solve $\frac{dy}{dx} = 12x$?

By Theorem 5.1, the derivative of a constant times x^2 scales by that constant, and a constant term's derivative is 0 — compare the result to the right-hand side of the equation. Final Answer: Yes — $\frac{dy}{dx} = 12x$, matching the equation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Addition and scalar multiplication

Let A be the matrix with rows $\langle -8, 4\rangle$ and $\langle 7, 7\rangle$, and B be the matrix with rows $\langle -2, 4\rangle$ and $\langle -7, -3\rangle$. Find $A + B$.

a. $\langle -8, 8\rangle, \langle 0, 4\rangle$

b. $\langle -10, 8\rangle, \langle 0, 2\rangle$

c. $\langle -10, 8\rangle, \langle 0, 4\rangle$ ✓

d. $\langle 0, 4\rangle, \langle -10, 8\rangle$

$A + B$ has rows $\langle -10, 8\rangle$ and $\langle 0, 4\rangle$.

22 [Foundational] • Target: Matrix multiplication

Let A be the matrix with rows $\langle -8, -2\rangle$ and $\langle 8, 8\rangle$, and B be the matrix with rows $\langle -4, -8\rangle$ and $\langle -5, 5\rangle$. Find $A + B$.

a. $\langle 3, 13\rangle, \langle -12, -10\rangle$

b. $\langle -10, -10\rangle, \langle 3, 13\rangle$

c. $\langle -12, -10\rangle, \langle 3, 13\rangle$ ✓

d. $\langle -12, -10\rangle, \langle 3, 11\rangle$

$A + B$ has rows $\langle -12, -10\rangle$ and $\langle 3, 13\rangle$.

23 [Foundational] • Target: Determinant and inverse

Let A be the matrix with rows $\langle 1, -3\rangle$ and $\langle 7, 6\rangle$, and B be the matrix with rows $\langle 6, 2\rangle$ and $\langle -5, 1\rangle$. Find $A + B$.

a. $\langle 7, -1\rangle, \langle 2, 5\rangle$

b. $\langle 2, 7\rangle, \langle 7, -1\rangle$

c. $\langle 7, -1\rangle, \langle 2, 7\rangle$ ✓

d. $\langle 9, -1\rangle, \langle 2, 7\rangle$

$A + B$ has rows $\langle 7, -1\rangle$ and $\langle 2, 7\rangle$.

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24 [Foundational] • Target: Solving systems with the inverse matrix

Let A be the matrix with rows $\langle 6, -2\rangle$ and $\langle -8, -6\rangle$, and B be the matrix with rows $\langle -5, -4\rangle$ and $\langle 5, -6\rangle$. Find $A + B$.

a. $\langle 1, -6\rangle, \langle -3, -12\rangle$ ✓

b. $\langle 3, -6\rangle, \langle -3, -12\rangle$

c. $\langle -3, -12\rangle, \langle 1, -6\rangle$

d. $\langle 1, -6\rangle, \langle -3, -14\rangle$

$A + B$ has rows $\langle 1, -6\rangle$ and $\langle -3, -12\rangle$.

25 [Foundational] • Target: Vector magnitude and dot product

Let A be the matrix with rows $\langle -4, 3\rangle$ and $\langle -7, 1\rangle$, and B be the matrix with rows $\langle -2, 9\rangle$ and $\langle -5, 1\rangle$. Find $A + B$.

a. $\langle -4, 12\rangle, \langle -12, 2\rangle$

b. $\langle -6, 12\rangle, \langle -12, 0\rangle$

c. $\langle -12, 2\rangle, \langle -6, 12\rangle$

d. $\langle -6, 12\rangle, \langle -12, 2\rangle$ ✓

$A + B$ has rows $\langle -6, 12\rangle$ and $\langle -12, 2\rangle$.

26 [Foundational] • Target: Angle between vectors

Let A be the matrix with rows $\langle 8, -7\rangle$ and $\langle 9, -7\rangle$, and B be the matrix with rows $\langle 7, 9\rangle$ and $\langle 1, 9\rangle$. Find $A + B$.

a. $\langle 17, 2\rangle, \langle 10, 2\rangle$

b. $\langle 10, 2\rangle, \langle 15, 2\rangle$

c. $\langle 15, 2\rangle, \langle 10, 2\rangle$ ✓

d. $\langle 15, 2\rangle, \langle 10, 0\rangle$

$A + B$ has rows $\langle 15, 2\rangle$ and $\langle 10, 2\rangle$.

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27 [Foundational] • Target: Matrix Inversion & Systems

Let A be the matrix with rows $\langle -6, 5\rangle$ and $\langle -9, 1\rangle$, and B be the matrix with rows $\langle 8, 3\rangle$ and $\langle 7, 6\rangle$. Find $A + B$.

a. $\langle 4, 8\rangle, \langle -2, 7\rangle$

b. $\langle 2, 8\rangle, \langle -2, 5\rangle$

c. $\langle 2, 8\rangle, \langle -2, 7\rangle$ ✓

d. $\langle -2, 7\rangle, \langle 2, 8\rangle$

$A + B$ has rows $\langle 2, 8\rangle$ and $\langle -2, 7\rangle$.

28 [Foundational] • Target: Determinants & Properties

Let A have rows $\langle -9, 8\rangle$ and $\langle -9, 6\rangle$, and let $\mathbf{v} = \langle 6, 3\rangle$. Find $A\mathbf{v}$.

a. $\langle -36, -30\rangle$

b. $\langle -28, -36\rangle$

c. $\langle -30, -36\rangle$ ✓

d. $\langle -30, -38\rangle$

$A\mathbf{v} = \langle (-9)(6) + (8)(3), (-9)(6) + (6)(3)\rangle = \langle -30, -36\rangle$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Dot Product & Projections

Let A have rows $\langle 9, -1\rangle$ and $\langle 4, -1\rangle$, and let $\mathbf{v} = \langle 3, -5\rangle$. Find $A\mathbf{v}$.

a. $\langle 17, 32\rangle$

b. $\langle 32, 15\rangle$

c. $\langle 34, 17\rangle$

d. $\langle 32, 17\rangle$ ✓

$A\mathbf{v} = \langle (9)(3) + (-1)(-5), (4)(3) + (-1)(-5)\rangle = \langle 32, 17\rangle$.

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30 [Intermediate] • Target: 3D Vectors & Cross Products

Let A have rows $\langle -4, 7\rangle$ and $\langle 6, 4\rangle$, and let $\mathbf{v} = \langle -8, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 69, -28\rangle$

b. $\langle 67, -28\rangle$ ✓

c. $\langle -28, 67\rangle$

d. $\langle 67, -30\rangle$

$A\mathbf{v} = \langle (-4)(-8) + (7)(5),\ (6)(-8) + (4)(5)\rangle = \langle 67, -28\rangle$.

31 [Intermediate] • Target: Addition and scalar multiplication

Let A have rows $\langle -2, 2\rangle$ and $\langle 8, -1\rangle$, and let $\mathbf{v} = \langle -9, -1\rangle$. Find $A\mathbf{v}$.

a. $\langle 18, -71\rangle$

b. $\langle -71, 16\rangle$

c. $\langle 16, -73\rangle$

d. $\langle 16, -71\rangle$ ✓

$A\mathbf{v} = \langle (-2)(-9) + (2)(-1),\ (8)(-9) + (-1)(-1)\rangle = \langle 16, -71\rangle$.

32 [Intermediate] • Target: Matrix multiplication

Let A have rows $\langle 4, -6\rangle$ and $\langle 8, -7\rangle$, and let $\mathbf{v} = \langle 6, -9\rangle$. Find $A\mathbf{v}$.

a. $\langle 111, 78\rangle$

b. $\langle 78, 111\rangle$ ✓

c. $\langle 80, 111\rangle$

d. $\langle 78, 109\rangle$

$A\mathbf{v} = \langle (4)(6) + (-6)(-9),\ (8)(6) + (-7)(-9)\rangle = \langle 78, 111\rangle$.

33 [Intermediate] • Target: Determinant and inverse

Let A have rows $\langle -3, 4\rangle$ and $\langle -1, -6\rangle$, and let $\mathbf{v} = \langle 1, -4\rangle$. Find $A\mathbf{v}$.

a. $\langle -19, 21\rangle$

b. $\langle -17, 23\rangle$

c. $\langle -19, 23\rangle$ ✓

d. $\langle 23, -19\rangle$

$A\mathbf{v} = \langle (-3)(1) + (4)(-4),\ (-1)(1) + (-6)(-4)\rangle = \langle -19, 23\rangle$.

34 [Intermediate] • Target: Solving systems with the inverse matrix

Let A have rows $\langle 2, -6\rangle$ and $\langle 3, 3\rangle$, and let $\mathbf{v} = \langle 3, 8\rangle$. Find $A\mathbf{v}$.

a. $\langle -42, 31\rangle$

b. $\langle -42, 33\rangle$ ✓

c. $\langle 33, -42\rangle$

d. $\langle -40, 33\rangle$

$A\mathbf{v} = \langle (2)(3) + (-6)(8),\ (3)(3) + (3)(8)\rangle = \langle -42, 33\rangle$.

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35 [Intermediate] • Target: Vector magnitude and dot product

Find $\det A$ for the matrix A with rows $\langle -5, -3\rangle$ and $\langle 6, 4\rangle$.

a. -2 ✓

b. 2

c. -4

d. 0

$\det A = (-5)(4) - (-3)(6) = -20 - (-18) = -2$, so A is invertible since $\det A \neq 0$.

36 [Intermediate] • Target: Angle between vectors

Find $\det A$ for the matrix A with rows $\langle -3, -3\rangle$ and $\langle -3, 5\rangle$.

a. -26

b. 24

c. -22

d. -24 ✓

$\det A = (-3)(5) - (-3)(-3) = -15 - 9 = -24$, so A is invertible since $\det A \neq 0$.

37 [Intermediate] • Target: Matrix Inversion & Systems

For $\mathbf{u} = \langle -20, -21\rangle$ and $\mathbf{w} = \langle -8, 7\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $15, 29$

b. $13, 31$

c. $29, 13$

d. $13, 29$ ✓

$\mathbf{u} \cdot \mathbf{w} = (-20)(-8) + (-21)(7) = 13$, and $|\mathbf{u}| = \sqrt{(-20)^2 + (-21)^2} = 29$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

For $\mathbf{u} = \langle -12, 35\rangle$ and $\mathbf{w} = \langle -7, 3\rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $191, 37$

b. $189, 37$ ✓

c. $37, 189$

d. $189, 39$

$\mathbf{u} \cdot \mathbf{w} = (-12)(-7) + (35)(3) = 189$, and $|\mathbf{u}| = \sqrt{(-12)^2 + 35^2} = 37$.

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39 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

Find the angle between $\langle -5, 5\rangle$ and $\langle 5, 2\rangle$, rounded to the nearest tenth of a degree.

a. 123.2°

b. 113.2° ✓

c. 108.2°

d. 118.2°

Dot product $= -15$, magnitudes $\sqrt{50}$ and $\sqrt{29}$. $\cos\theta = \frac{-15}{\sqrt{1450}} \approx -0.3939 \Rightarrow \theta \approx 113.2^\circ$.

40 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Find the angle between $\langle 4, 6\rangle$ and $\langle -6, 1\rangle$, rounded to the nearest tenth of a degree.

a. 124.2°

b. 119.2°

c. 114.2° ✓

d. 109.2°

Dot product $= -18$, magnitudes $\sqrt{52}$ and $\sqrt{37}$. $\cos\theta = \frac{-18}{\sqrt{1924}} \approx -0.4104 \Rightarrow \theta \approx 114.2^\circ$.

41 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 5.01 Introductory Calculus

What is the inverse of $A = [[4, 7], [1, 2]]$?

a. $[[2, -7], [-1, 4]]$ ✓

b. $[[-2, 7], [1, -4]]$

c. $[[4, -7], [-1, 2]]$

d. $[[2, 1], [7, 4]]$

$\det(A) = 4(2) - 7(1) = 8 - 7 = 1$. Inverse is $\frac{1}{1}[[d, -b], [-c, a]] = [[2, -7], [-1, 4]]$.

42 [Challenging] • Target: Matrix Inversion & Systems • Cross-Lesson: 6.01 Probability

If $AX = B$, where A is invertible, which expression gives the solution column vector X?

a. $A^{-1}B$ ✓

b. BA^{-1}

c. B/A

d. AB^{-1}

Multiply on the left by A^{-1}: $A^{-1}AX = A^{-1}B \Rightarrow X = A^{-1}B$.

43 [Challenging] • Target: Determinants & Properties • Cross-Lesson: 1.01 Complex Numbers

What happens to the determinant of a matrix if two of its rows are interchanged?

a. **Its sign is reversed** ✓

b. It doubles

c. It remains unchanged

d. It becomes zero

Interchanging any two rows of a square matrix multiplies its determinant by -1.

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44 [Challenging] • Target: Dot Product & Projections • Cross-Lesson: 2.01 Sequences & Series

What is the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$?

a. $\langle 3, 0 \rangle$ ✓

b. $\langle 4, 0 \rangle$

c. $\langle 0, 4 \rangle$

d. $\langle 1, 0 \rangle$

$\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\mathbf{v} = \frac{3(1) + 4(0)}{1^2}\langle 1, 0 \rangle = 3\langle 1, 0 \rangle = \langle 3, 0 \rangle$.

45 [Challenging] • Target: 3D Vectors & Cross Products • Cross-Lesson: 3.01 Functions & Graphs

Compute the cross product $\langle 1, 0, 0 \rangle \times \langle 0, 1, 0 \rangle$.

a. $\langle 0, 0, 1 \rangle$ ✓

b. $\langle 0, 0, -1 \rangle$

c. $\langle 1, 1, 0 \rangle$

d. $\mathbf{0}$

$\mathbf{i} \times \mathbf{j} = \mathbf{k} = \langle 0, 0, 1 \rangle$.

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